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FIFA Ranking Points and Recent Football National Team Wins: A Bayesian Linear Regression Analysis

1. Introduction

In international football, FIFA points are the ranking points used by the International Federation of Association Football to compare national teams. The FIFA ranking system adds or subtracts points after international matches, depending on the match result and also on the strength of the opponent (Fédération Internationale de Football Association [FIFA], 2024). Since the 2026 World Cup starts this month of June 2026, these rankings are especially relevant because they are often used to judge which teams are expected to be stronger and potentially go further in the tournament rounds (FIFA, 2026). This is what motivated the idea of doing this report.

However, FIFA points are also partly based on past match results. For that reason, I do not treat this report as a causal analysis. I am not asking whether FIFA points cause teams to win more matches. Instead, the question is more descriptive: do teams with higher pre-tournament FIFA points also have more wins in the four years before the World Cup? In other words, I use this model to study how well FIFA points reflect recent winning volume.

The outcome is the number of wins in the previous four years, and the predictor is standardized FIFA points before the tournament. The model is:

$$wins_i = \alpha + \beta x_i + \varepsilon_i, \qquad \varepsilon_i \sim N(0, \sigma^2),$$

where x_i is standardized FIFA points. The parameter of interest is β. Since the predictor is standardized, β is the expected change in wins for a one-standard-deviation increase in FIFA points. The main substantive hypothesis is:

$$H_1: \beta > 0.$$

The Bayesian procedure is appropriate here for three reasons. First, it gives the full posterior distribution of β, not only a point estimate. Second, it lets me directly calculate $P(\beta > 0 \mid \text{data})$, which is exactly linked to the research question. Third, the Normal linear regression model with Normal priors and an Inverse-Gamma prior for σ^2 has closed-form conditional posteriors, so I can estimate it with a hand-coded Gibbs sampler instead of relying on pre-packaged estimation functions. For Bayesian testing, I use a posterior predictive Bayesian p-value under the null model $H_0: \beta = 0$.

2. Data

The data used in this report is the FIFA World Cup Team Dataset available on Kaggle (Harashi, 2026). This dataset contains team-level information for World Cup tournaments from 2002 to 2026. Each row represents one national team before a specific World Cup, so the data is useful for comparing teams based on pre-tournament characteristics and recent performance. For this report, I only use the variables that are directly linked to my research question: the number of wins in the four years before the World Cup, wins_last_4y, and the FIFA points before the tournament, fifa_points_pre_tournament.

The dataset is publicly available on Kaggle, and is shared under the Creative Commons Attribution 4.0 International license. This means that the data can be reused and adapted, as long as proper credit is given to the original creator (Creative Commons, n.d.).

I standardized FIFA points using: $x_i = \frac{FIFA_i - \overline{FIFA}}{s_{FIFA}}$.

This makes the interpretation of the slope easier, because β becomes the expected change in wins for a one-standard-deviation increase in FIFA points.

The dataset used in the model has $n = 192$ observations. The average number of wins in the last four years is about 26.05, with a standard deviation of about 6.91. The mean FIFA-points value is about 954.17, with a standard deviation of about 367.54.

Table 1: Descriptive statistics

Quantity	Value
n	192.000
Mean wins	26.047
SD wins	6.913
Mean FIFA points	954.165
SD FIFA points	367.545
Correlation	0.325
Shapiro-Wilk p	0.038

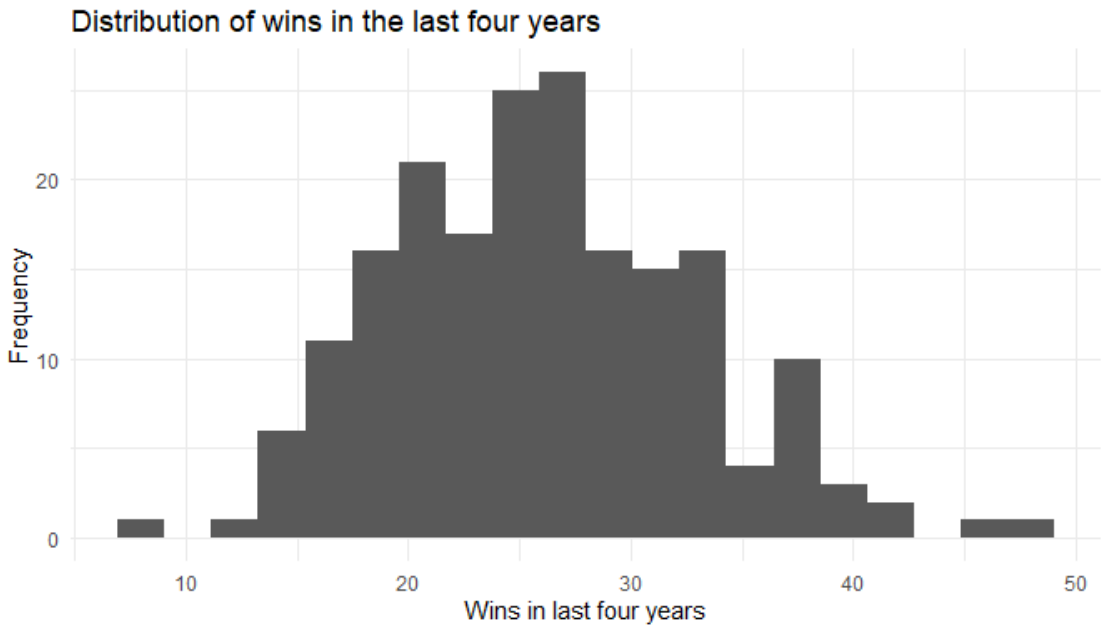


Figure 2: Histogram of wins in the last four years.

The histogram shows that the wins variable is roughly centered around the mid-20s. The Shapiro-Wilk test gives: $p = 0.038$. At the 5% cut-off level, this suggests that the outcome is not perfectly normally distributed. However, the p-value is close to 0.05, and the distribution is not extremely non-normal, and add to that the mean of the count variable is relatively large, I decided to treat the Normal model as a reasonable approximation. I will discuss this again in the limitations section.

3. Bayesian model, priors, and Gibbs sampler

The likelihood is:

$$y_i \mid \alpha, \beta, \sigma^2 \sim N(\alpha + \beta x_i, \sigma^2).$$

I use the following priors:

$$\alpha \sim N(25, 10^2), \qquad \beta \sim N(0, 5^2), \qquad \sigma^2 \sim IG(2,49).$$

The prior for α is centered at 25 because the average number of wins is around that number. The prior standard deviation is 10, which is wide enough to allow a broad range of plausible intercept values.

The prior for β is centered at zero. This is important because I do not want the prior to force the conclusion that FIFA points are positively associated with wins. Since FIFA points are standardized, a slope of 5 would already mean that a team one standard deviation above the mean FIFA-points value is expected to have 5 more wins. In football terms, this is already a large effect, so $N(0, 5^2)$ is still a fairly broad prior. The prior for σ^2 puts the residual variance on approximately the same scale as the observed outcome. Since the standard deviation of wins is around 7, a variance around $7^2 = 49$ is reasonable as a prior scale. The prior is not meant to be very strong, but it prevents the variance from being completely unrealistic.

For the Gibbs sampler, define $\theta = (\alpha, \beta)'$ and $y \sim N(X\theta, \sigma^2 I)$. The prior is $\theta \sim N(m_0, V_0)$. The conditional posterior for the regression coefficients is:

$$\theta \mid \sigma^2, y \sim N(m_n, V_n),$$

where:

$$V_n = \left(V_0^{-1} + \frac{1}{\sigma^2} X'X\right)^{-1}, \qquad m_n = V_n \left(V_0^{-1} m_0 + \frac{1}{\sigma^2} X'y\right).$$

The conditional posterior for the variance is:

$$\sigma^2 \mid \theta, y \sim IG\left(a_0 + \frac{n}{2},\ b_0 + \frac{1}{2}\sum_{i=1}^{n}(y_i - \alpha - \beta x_i)^2\right).$$

In each iteration I sample θ from its Normal conditional posterior and then sample σ^2 from its Inverse-Gamma conditional posterior. I ran three chains with 10,000 iterations and discarded the first 2,000 as burn-in.

4. Convergence and Bayesian estimation results

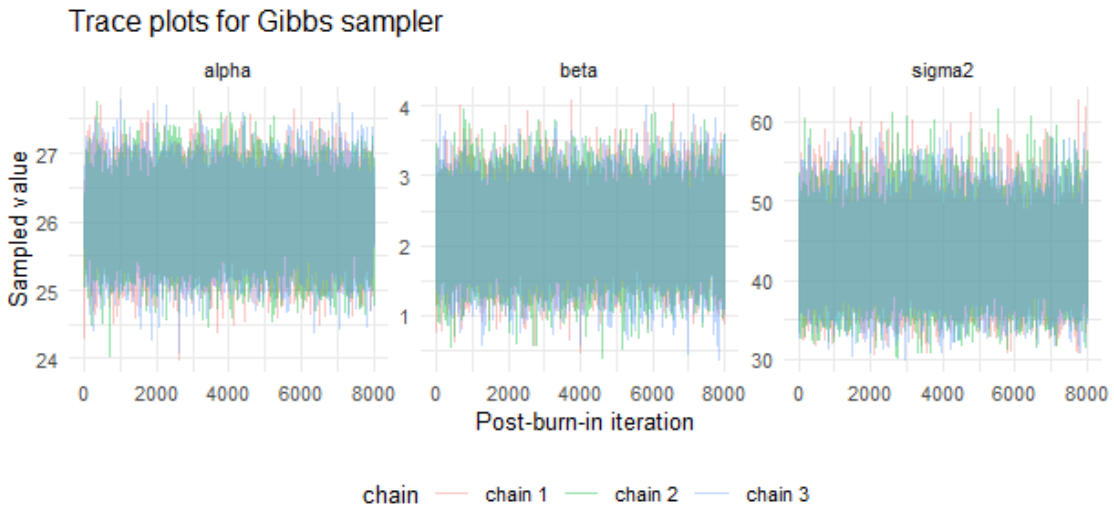


Figure 3: Trace plots for the Gibbs sampler.

The trace plots show no clear trend and the three chains overlap well forming a hairy fat caterpillar like figure. This suggests that the chains are sampling from the same posterior distribution. The $\hat{R}$ values are approximately 1 and the effective sample sizes are large, which supports convergence.

Table 3: Posterior summaries and convergence diagnostics.

Parameter	EAP	PSD	95% CCI	Rhat	ESS
alpha	26.04	0.47	[25.12, 26.96]	1	24292
beta	2.23	0.47	[1.3, 3.14]	1	23724
sigma2	42.97	4.37	[35.21, 52.31]	1	24000

The posterior mean of β is 2.23. This means that a team one standard deviation above average FIFA points is expected to have about 2.23 more wins in the previous four years. The 95% credible interval is $[1.3,3.14]$, which is fully above zero. The posterior probability of the main hypothesis is:

$$P(\beta > 0 \mid \text{data}) > .999.$$

This is more informative than just saying that the slope is significant. It says that, after combining the prior and the likelihood (data), practically all posterior mass for β is above zero.

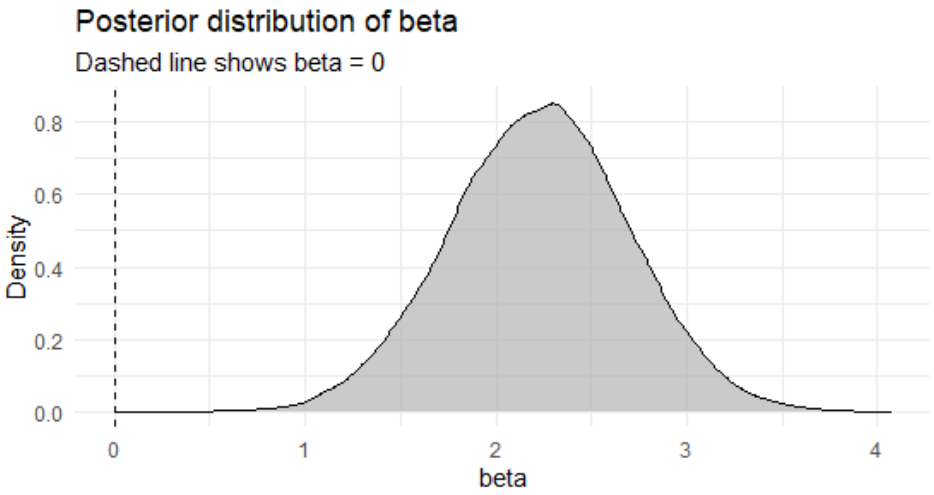


Figure 4: Posterior distribution of beta.

The posterior distribution is clearly away from the prior centre of zero. I believe this is important because I did not choose a prior that already forced the result to be positive. The prior still allowed negative slopes, values close to zero, and positive slopes. After using the data, however, most of the posterior distribution is concentrated on positive values of β. This makes me more confident that the positive relationship comes mainly from the data, and not from a prior that already assumed the answer. I also return to this point in the next section, where I check the sensitivity of the results to different prior settings.

5. Bayesian p-value and sensitivity to prior information

For Bayesian testing, I use a posterior predictive check under: $H_0: \beta = 0$. Under this null model, FIFA points do not help explain wins. I fit an intercept-only Bayesian model and simulate replicated datasets from it. For each replicated dataset I calculate the same statistic as in the observed data: $T(y) = cor(x, y)$. I chose this statistic because the hypothesis is about a positive association between FIFA points and wins. In this regression, the correlation captures the same main pattern as the slope β. I generated 5,000 replicated datasets from the null model after discarding 2,000 burn-in iterations, and computed $cor(x, y^{rep})$for each one.

The observed value is $T_{obs} = 0.325$. The Bayesian p-value is: $p_B = P(T(y^{rep}) \geq T_{obs} \mid \text{data}, H_0)$.

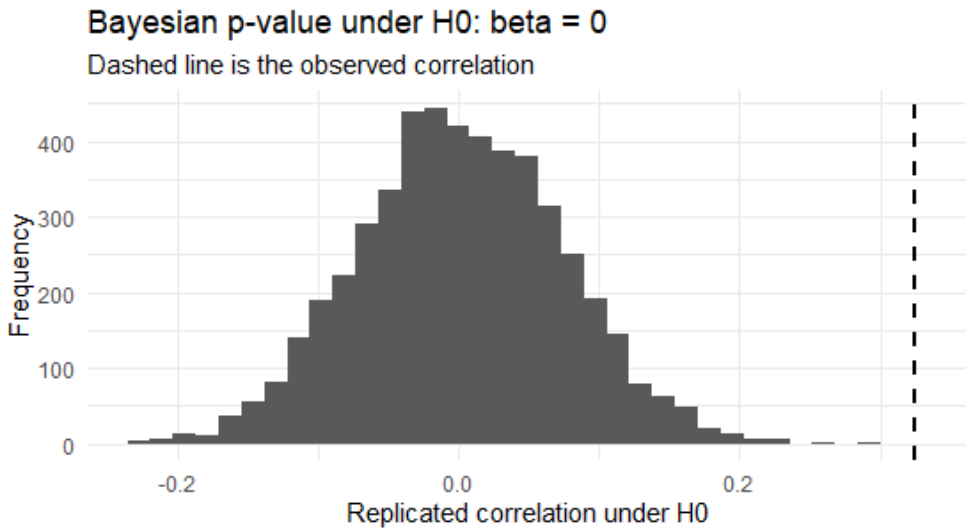


Figure 5: Posterior predictive Bayesian p-value under the null model.

The observed correlation is far in the right tail of the replicated correlations. **The resulting Bayesian p-value is $\mathbf{1.9996 \times 10^{-4}}$.** This indicates that datasets generated under the null model rarely produce a correlation as large as the one observed. Therefore, the null model ($\beta = 0$) does not appear to reproduce this feature of the data well. Because the Bayesian p-value is a posterior predictive model-checking measure rather than a direct probability that the null hypothesis is false, it should not be interpreted as an exact decision rule. Nevertheless, the very small value suggests a substantial discrepancy between the observed data and the predictions of the null model.

For sensitivity analysis, I vary the amount of prior information for β using fractions on information parameter J, $2J$, and $3J$. The prior remains centered at the null value $\beta = 0$, but the prior standard deviation changes as:

$$\sigma_{\beta,J} = \sqrt{\frac{n}{J}} \times SE_{OLS}(\hat{\beta}).$$

Larger values of Jcorrespond to increasingly informative priors because they reduce the prior standard deviation and concentrate more probability mass around $\beta = 0$. This checks whether the posterior conclusion depends on one arbitrary prior scale.

Table 4: Prior sensitivity for beta using J, 2J, and 3J.

Prior	J	Prior SD	EAP beta	95% CCI	P(beta > 0)
J	1	6.57	2.24	[1.31, 3.17]	1
2J	2	4.65	2.22	[1.28, 3.16]	1
3J	3	3.79	2.21	[1.28, 3.1]	1

The posterior mean of β remains around 2.2 under J, $2J$, and $3J$, and the credible interval stays above zero. Therefore, the conclusion is stable under different amounts of prior information.

6. Classical frequentist comparison and residual check

The frequentist version of what I did is the same linear regression estimated with ordinary least squares. It gives the same substantive direction: the slope is positive and the one-sided p-value is very small.

Table 5: Bayesian and classical comparison.

Analysis	Quantity	Result
Bayesian posterior	Posterior mean beta	2.227
Bayesian posterior	P(beta > 0 | data)	> .999
Bayesian p-value	Observed correlation	0.325
Bayesian p-value	p_B under H0: beta = 0	2e-04
Classical OLS	OLS slope	2.246
Classical OLS	One-sided p-value	2.13e-06
Classical OLS	R-squared	0.106

The classical result answers a different kind of question. The p-value is about how unusual the observed slope would be in repeated samples if the true slope were zero. The Bayesian analysis instead gives a posterior distribution for β and allows a direct statement about $P(\beta > 0 \mid \text{data})$. For this research question, I find the Bayesian interpretation more natural because the question is directly about uncertainty in the slope itself. At the same time, the classical model is useful as a check, because it leads to the same broad conclusion.

The classical R^2 is 0.106, so FIFA points explain about 10.6% of the variation in recent wins. This is enough to show a real association, but it also shows that FIFA points alone are incomplete.

As a small extension, I use the posterior mean regression line to identify teams that overperformed or underperformed the model expectation. I compute:

$$residual_i = wins_i - \widehat{wins}_i.$$

Positive residuals mean that a team won more matches often than expected from FIFA points alone, while negative residuals mean that a team won less matches often than expected.

Table 6: Selected teams that overperformed or underperformed the model expectation.

Type	Team and year	Observed wins	Predicted wins	Residual wins
Overperformed	Saudi Arabia 2002	48	24.06	23.94
Overperformed	Mexico 2002	41	24.60	16.40
Overperformed	Mexico 2010	42	25.69	16.31

Overperformed	Spain 2010	46	29.74	16.26
Underperformed	New Zealand 2010	8	22.75	-14.75
Underperformed	Ghana 2022	16	28.70	-12.70
Underperformed	Slovenia 2010	14	25.47	-11.47
Underperformed	Ecuador 2022	18	29.13	-11.13

I decided to add this part as a football interpretation to the residuals. It shows that FIFA points are useful, but there are still teams whose recent win volume is higher or lower than what the one-predictor model expects.

7. Reflection and conclusion

The Bayesian estimation gives a clear substantive result. The posterior mean of β is about 2.23, with a 95% credible interval from 1.3 to 3.14. Since the full interval is above zero and $P(\beta > 0 \mid \text{data}) > .999$, the posterior as a whole supports a positive association between FIFA points and recent wins for the national teams. The sensitivity analysis also matters here: even when I changed the amount of prior information using J, $2J$, and $3J$, the posterior conclusion stayed the same, which demonstrates that the result is not mainly a consequence of one specific prior choice.

The Bayesian p-value supports the same conclusion from a model-checking perspective. Under the null model $H_0: \beta = 0$, replicated datasets almost never produced a correlation as large as the observed one. This does not mean the Bayesian p-value is identical to a classical p-value; I interpret it as showing that the no-slope model does not reproduce the observed association well.

Compared with the classical frequentist approach, the Bayesian approach is more transparent about uncertainty in the parameter. The classical model gives an OLS estimate, confidence interval, and p-value. These are useful, but their interpretation depends on repeated-sampling logic. The Bayesian model gives a posterior distribution and makes it possible to say directly how much posterior probability supports the positive slope. It also forces me to state the prior assumptions clearly and check whether the conclusion is sensitive to them.

There are also limitations. First, wins_last_4y is a count variable, while I model it using Normal linear regression. This is a mere simplification for the sake of this report, because a Poisson or negative binomial model would be more natural for a fully count-based analysis. However, in this case, the Normal model is still acceptable as the histogram of wins_last_4y does not show a strongly skewed. Second, FIFA points and wins are conceptually related because both are based on match results. Therefore, this is not a causal analysis. Third, the model uses only one predictor while other covariates like number of matches played could potentially improve the model.

Overall, I conclude that FIFA points do reflect recent win volume, but only partly. Higher FIFA points are clearly associated with more recent wins, but the modest R^2 and the residual analysis show that FIFA points are far from a complete measure of team performance. The main value of the Bayesian analysis is that it gives a full posterior interpretation of this association and shows that the conclusion is stable across prior sensitivity checks.

8. References

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9. AI Statement

I used ChatGPT 5.5 as a support tool during the preparation of this report. I mainly used it to help me debug parts of the R code, improve the structure and readability of the code, refine some graphs, and polish the wording of some paragraphs. The main ideas, text, procedure, statistical steps, interpretation, and final decisions were my own and were based on the Bayesian Statistics course material.

Examples of prompts I used include:

- “Is there a strong rationale for using a Bayesian p-value in this analysis?”
- “How can I make this graph look clearer and more professional?”
- “Can you make this sound clearer while keeping my own argument?”

I used the AI output critically and did not copy results without checking them against my code, the data, and the course material.